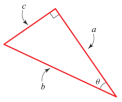
5.2

1)



a) $\boxed{a: \text{adjacent}}$

$\boxed{b: \text{hypotenuse}}$

$\boxed{c: \text{opposite}}$

b) trig functions

$\sin\theta = \boxed{\dfrac{\text{opposite}}{\text{hypotenuse}}}$

$\cos\theta = \boxed{\dfrac{\text{adjacent}}{\text{hypotenuse}}}$

$\tan\theta = \boxed{\dfrac{\text{opposite}}{\text{adjacent}}}$

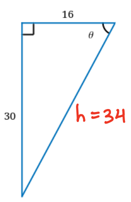
2) reciprocal identities

$\csc\theta = \boxed{\dfrac{1}{\sin\theta}}$

$\sec\theta = \boxed{\dfrac{1}{\cos\theta}}$

$\cot\theta = \boxed{\dfrac{1}{\tan\theta}}$

3) find exact values of 6 trig ratios



$h = \sqrt{16^2 + 30^2}$
$= \sqrt{1156}$
$= 34$

$\sin\theta = \boxed{\dfrac{30}{34}}$

$\cos\theta = \boxed{\dfrac{16}{34}}$

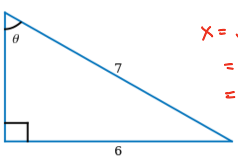
$\tan\theta = \boxed{\dfrac{30}{16}}$

$\csc\theta = \boxed{\dfrac{34}{30}}$

$\sec\theta = \boxed{\dfrac{34}{16}}$

$\cot\theta = \boxed{\dfrac{16}{30}}$

4) find exact values of 6 trig ratios



$x = \sqrt{13}$

$x = \sqrt{7^2 - 6^2}$
$= \sqrt{49 - 36}$
$= \sqrt{13}$

$\sin\theta = \boxed{\dfrac{6}{7}}$

$\cos\theta = \boxed{\dfrac{\sqrt{13}}{7}}$

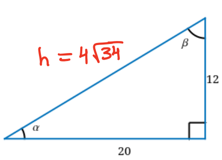
$\tan\theta = \dfrac{6}{\sqrt{13}} = \boxed{\dfrac{6\sqrt{13}}{13}}$

$\csc\theta = \boxed{\dfrac{7}{6}}$

$\sec\theta = \dfrac{7}{\sqrt{13}} = \boxed{\dfrac{7\sqrt{13}}{13}}$

$\cot\theta = \boxed{\dfrac{\sqrt{13}}{6}}$

5) find $\sin(\alpha)$, $\cos(\beta)$, $\tan(\alpha)$, $\cot(\beta)$, $\sec(\alpha)$, $\csc(\beta)$



$h = 4\sqrt{34}$

$h = \sqrt{20^2 + 12^2}$
$= \sqrt{544}$
$= 4\sqrt{34}$

a) $\sin(\alpha)$ and $\cos(\beta)$

$$\frac{12}{4\sqrt{34}}$$
$$= \frac{3}{\sqrt{34}}$$
$$= \boxed{\frac{3\sqrt{34}}{34}}$$

b) $\tan(\alpha)$ and $\cot(\beta)$

$$= \boxed{\frac{12}{20}}$$

c) $\sec(\alpha)$ and $\csc(\beta)$

$$= \frac{4\sqrt{34}}{20}$$
$$= \boxed{\frac{\sqrt{34}}{5}}$$

6) use calculator to evaluate

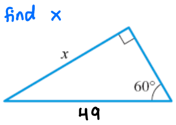
a) $\sec(3)$

$$= \frac{1}{\cos(3)}$$
$$\boxed{\approx -1.01011}$$

b) $\tan(58°)$

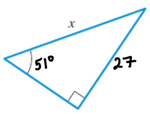
$$\boxed{\approx 1.60033}$$

7) find x



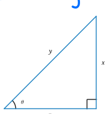
$$\sin(60°) = \frac{x}{49}$$
$$= 49\sin(60°)$$
$$\boxed{x = \frac{49\sqrt{3}}{2}}$$

8) find x rounded to 5 decimal places



$$\sin(51°) = \frac{27}{x}$$
$$x = \frac{27}{\sin(51°)}$$
$$\boxed{x \approx 34.74251}$$

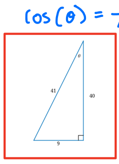
9) express x and y in terms of trig ratios of θ



$$\tan\theta = \frac{x}{7}$$
$$\boxed{x = 7\tan\theta}$$
$$\cos\theta = \frac{7}{y}$$
$$\boxed{y = \frac{7}{\cos\theta}}$$

10) sketch a triangle that has acute angle θ

$$\cos(\theta) = \frac{40}{41}$$



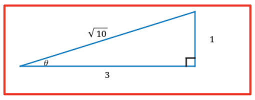
find other 5 trig ratios of θ

$$\sin\theta = \boxed{\frac{9}{41}}$$
$$\tan\theta = \boxed{\frac{9}{40}}$$
$$\csc\theta = \boxed{\frac{41}{9}}$$
$$\sec\theta = \boxed{\frac{41}{40}}$$

$\cot\theta = \boxed{\frac{40}{9}}$

11) sketch a triangle that has acute angle θ

$\cot(\theta) = 3$



find other 5 trig ratios of θ

$\sin\theta = \frac{1}{\sqrt{10}} = \boxed{\frac{\sqrt{10}}{10}}$

$\cos\theta = \frac{3}{\sqrt{10}} = \boxed{\frac{3\sqrt{10}}{10}}$

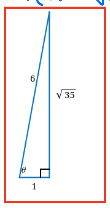
$\csc\theta = \frac{\sqrt{10}}{1} = \boxed{\sqrt{10}}$

$\sec\theta = \boxed{\frac{\sqrt{10}}{3}}$

$\tan\theta = \boxed{\frac{1}{3}}$

12) sketch a triangle that has acute angle θ

$\tan(\theta) = \sqrt{35}$



find other 5 trig ratios of θ

$\sin\theta = \boxed{\frac{\sqrt{35}}{6}}$

$\cos\theta = \boxed{\frac{1}{6}}$

$\csc\theta = \frac{6}{\sqrt{35}} = \boxed{\frac{6\sqrt{35}}{35}}$

$\sec\theta = \frac{6}{1} = \boxed{6}$

$\cot\theta = \frac{1}{\sqrt{35}} = \boxed{\frac{\sqrt{35}}{35}}$

13) evaluate without calculator

$\sin\left(\frac{\pi}{4}\right) + \cos\left(\frac{\pi}{4}\right)$

$= \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}$

$= \boxed{\sqrt{2}}$

14) evaluate without calculator

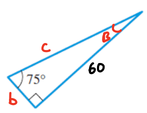
$(\sin(30°))^2 + (\cos(30°))^2$

$= \left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2$

$= \frac{1}{4} + \frac{3}{4}$

$= \boxed{1}$

15) solve triangle below



length of side adjacent to given angle

$\tan(75°) = \frac{60}{b}$

$b = \frac{60}{\tan(75°)}$

$\boxed{\approx 16.08 \text{ un}}$

length of hypotenuse

$\sin(75°) = \frac{60}{c}$

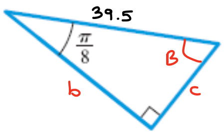
$c = \frac{60}{\sin(75°)}$

$\boxed{\approx 62.12 \text{ un}}$

other acute angle

$m\angle B = 90° - 75° = \boxed{15°}$

16) solve triangle below



length of side opposite to given angle

$$\sin\left(\frac{\pi}{8}\right) = \frac{c}{39.5}$$

$$c = 39.5 \sin\left(\frac{\pi}{8}\right)$$

$$c \boxed{\approx 15.12 \text{ un}}$$

length of side adjacent to given angle

$$\cos\left(\frac{\pi}{8}\right) = \frac{b}{39.5}$$

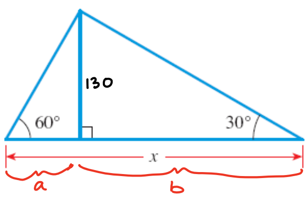
$$b = 39.5 \cos\left(\frac{\pi}{8}\right)$$

$$b \boxed{\approx 36.49 \text{ un}}$$

other acute angle

$$m\angle B = \frac{\pi}{2} - \frac{\pi}{8} = \boxed{\frac{3\pi}{8} \text{ rad}}$$

17) find x



$$x = a + b$$

$$\tan(30°) = \frac{130}{b}$$

$$b = \frac{130}{\tan(30°)}$$

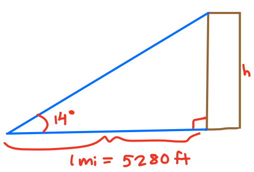
$$\tan(60°) = \frac{130}{a}$$

$$a = \frac{130}{\tan(60°)}$$

$$x = \frac{130}{\tan(30°)} + \frac{130}{\tan(60°)}$$

$$\boxed{x \approx 300.2 \text{ un}}$$

18) angle of elevation to top of a very tall building is 14° from ground at a distance of one mile from base of building. find height of building in feet.



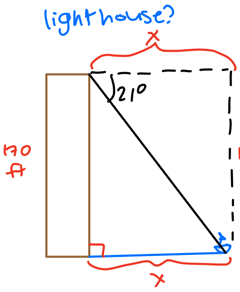
$$\tan(14°) = \frac{h}{5280}$$

$$h = 5280 \tan(14°)$$

$$h \approx 1316$$

$\boxed{\text{The height of the building is} \approx 1316 \text{ ft.}}$

19) from top of a 170-foot-tall lighthouse, angle of depression to a ship in the ocean is 21°. how far in ft is the ship from base of the lighthouse?



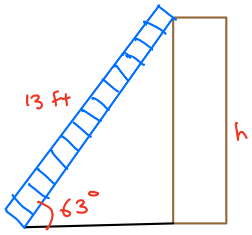
$$\tan(21°) = \frac{170}{x}$$

$$x = \frac{170}{\tan(21°)}$$

$$x \approx 443 \text{ ft}$$

$\boxed{\text{The ship is} \approx 443 \text{ ft from base of lighthouse.}}$

20) 13-foot-tall ladder leans against building so that the angle between the ground and ladder is 63°. How high in ft does the ladder reach on the building?



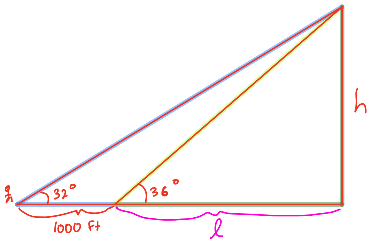
$\sin(63°) = \frac{h}{13}$

$h = 13\sin(63°)$

$h \approx 12$ ft

The ladder reaches ≈ 12 ft high on the building.

21) to estimate height of a mountain above a level plain, angle of elevation to the top of the mountain a surveyor measures to be 32°. One thousand feet closer to the mountain along the plain, angle of elevation is 36°. estimate height of mountain in feet.



Triangle 1

$\tan 32° = \frac{h}{\ell + 1000} \rightarrow \ell + 1000 = \frac{h}{\tan 32°}$

Triangle 2

$\tan 36° = \frac{h}{\ell} \rightarrow \ell = \frac{h}{\tan 36°}$

$\frac{h}{\tan 36°} + 1{,}000 = \frac{h}{\tan 32°}$

$1000 = \frac{h}{\tan 32°} - \frac{h}{\tan 36°}$

$1000 = h\left(\frac{1}{\tan 32°} - \frac{1}{\tan 36°}\right)$

$h = \frac{1000}{\frac{1}{\tan 32°} - \frac{1}{\tan 36°}}$

$h \approx 4465$ ft

The height of the mountain is ≈ 4465 ft.